

Co-Presence, Connection, and Co-Creation: Building Real-time Cross-person Neurofeedback Interactions

Building Real-time Cross-person Neurofeedback Interactions

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ABSTRACT

How can we push the boundaries of human connection by tapping into the power of immersive interpersonal brainwave interactions? While biosignal visualizations and interactions have advanced in both research and art scenes, most experiences and applications that utilize brainwaves and other biosignals involve single-person or one-way interactions from one individual to another, underexploiting the prosocial potential of these biosignals. This talk explores cross- and multi-person environments in virtual reality (VR) driven by shared brainwaves. Through case studies of two installations in a series, each showcasing unique relationships cultivated by the immersive experience via distinct EEG manifestations and visualization rules - feeling each other's presence, reinforcing mutual emotional connections, and co-creating art through synchronized brainwaves - the talk describes the design and development process of cross-person neurofeedback experiences in virtual reality. It explains various innovative approaches to real-time brainwave visualizations and interactions that integrate technology, science, storytelling and gamification, and examines their significance in expanding non-verbal channels of communication, elevating interpersonal connections, and fostering collective creativity.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Collaborative interaction**;

KEYWORDS

Generative VR, Interpersonal social connectedness, Co-creation, EEG

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1 INTRODUCTION AND RELATED WORK

We drew inspiration from a fascinating phenomenon known as "embodied simulation" or the "mirror effect." It's the natural tendency that we synchronize with others in various ways from matching footsteps during a stroll to mirroring gestures during conversations and syncing heart rates and breathing while watching emotional films together [McCraty, 2017]. Intrigued by this phenomenon, we were motivated to explore the potential of utilizing brainwave data to facilitate prosocial interactions in both cross- and multi-person settings.

Building upon previous research on utilizing biometric data, particularly EEG signals, for enhancing emotional connections and collaborative performance, several studies suggest that sharing brainwave data with others can foster social connectedness, intuitive understanding, and empathy [Hassib et al., 2017][Yun et al., 2012][Dumas et al., 2010]. These findings underscore the untapped potential of synchronizing interpersonal brainwaves in enhancing emotional experiences and collaborative endeavors. We also reference studies examining the influence of external visual stimuli on adjusting brainwave ratios to enhance focus [Horie et al., 2018], and highlighting the capacity for brainwaves to align with diverse external frequencies across visual, auditory, and tactile modalities [Will and Berg, 2007]. Within a series of case studies, we applied VR visual stimuli to enhance brainwave synchronization and foster emotional connections in the realms of co-creation and co-performance, providing a unique gamified and interactive experience.

2 DESIGN

2.1 SinkInSync

SinkInSync is a generative VR experience that uses one user's brainwave data to procedurally render 3D scenes and passively displays visual cues that are synchronized with the real-time brainwave frequency to another user [Wang and Feng, 2023]. This experience explores the potential of VR as an avenue for augmenting cognitive and emotional social connectedness in remote interactions via externally induced brainwave synchronization between pairs of individuals.

The visual elements within the scene provide users with an intuitive understanding and immersive experience of their partner's brainwave data. The colors help the user identify the dominant brainwave frequency and, consequently, the mental state of their partner user, while the pulsations visually represent the level of mental activity and facilitate the synchronization of mental rhythms between users. Furthermore, when users' brainwaves synchronize

within a particular band, ethereal night-blooming cereus of matching hues blossom beneath their feet momentarily before dissolving away. This lively spectacle commemorates their neural harmony and connection, forging memorable experiences that users eagerly await, where their brainwaves unite seamlessly.

2.2 SynCoreate

Unlike SinkInSync that explores brainwave synchronization and gamified experience, SynCoreate presents an engaging VR experience where users collaboratively craft artwork by synchronizing their brainwaves. The setting unfolds within a starry dome arena, intricately crafted to mirror the hemispherical contours of the brain, guiding users into an imaginative cognitive realm. Otherworldly sprites drift gracefully, each influenced by the real-time brainwave signals of a user. When the brainwaves of two users synchronize at a specific frequency, bursts of paint particles materialize on the dome's three-dimensional canvas, with colors corresponding to the synchronized brainwave type. With each synchronization event, the canvas becomes adorned with more paint, gradually forming a vibrant and distinct painting resembling a sky illuminated by fireworks. This evolving artwork celebrates the users' connection and provides a tangible manifestation of their shared cognitive experiences.

In this scene, the mapping of sprite and paint stroke positions is artistically interpreted from research findings that establish the relationship between brainwave bands and brain regions [Zheng et al., 2017] [Li et al., 2022], harmonized with symmetry rules, and adjusted to ensure overlaps among the areas, enabling paints to blend and coalesce into dynamic 'neural paintings'.

3 IMPLEMENTATION

Using Muse headbands, a commercially available brainwave sensing device, we gather real-time brainwave data from each user. The brainwave data, including dominant band types and frequencies, is processed and imported into Unity through an OSC server and the built-in WebSocket library. The brainwave data is categorized into five band types: Gamma (γ) (above 30 Hz), Beta (β) (14 - 29 Hz), Alpha (α) (8-13 Hz), Theta (θ) (4-7 Hz), and Delta (δ) (1-3 Hz). In the Unity environment, we map these real-time brainwave data from the users to CG properties within the scene and display it to the other user(s) for cross-person sync or collaboration, crafting a dynamic, cross-person visual representation of brain activities.

4 DISCUSSION AND FUTURE WORK

Playtesting with audiences at multiple exhibition venues showed the ability of SinkInSync to enhance interpersonal emotional awareness and foster a sense of togetherness without verbal communication, and SynCoreate to build mutual rapport and boost creativity. Based on user feedback, we've identified several key design considerations for crafting real-time cross-person EEG environments. **Co-presence nurtures connection and co-creation**, with users expressing desire to purposefully observe their counterpart's mental state, compare against their own in real-time, and synchronize collaboratively for celebratory moments. This underscores the value of cross- and multi-person design, emphasizing brainwave co-presence and interactions among users. **Intuitive and adaptive**

mapping rules are preferred, addressing the interest in internalizing brainwave classifications and color mapping for active pattern tracking. Unlike traditional data visualization, 2D legends and annotations may distract from immersive mindfulness experiences.

Balancing specificity and abstraction of the virtual environment is crucial, as environmental cues like pulsations, sound and specific object associations actively moderate biosignals. Being intentional and reductive about these cues enhances the synchronization process. **Data transparency through clear feedback loops reinforces the user experience and trust**. This includes setting goals for users to achieve, showcasing results through gamification, special effects, and collaborative artistic tasks, and enabling single-individual mode with past-self data for onboarding or solitary experiences.

This is a long-term project series that combines virtual platforms with biometric data synchronization, offering fresh perspectives on dynamic visualizations, interaction modalities, telepresence experiences, and interpersonal connections. Future plans include customizable mappings to enhance personalization, incorporating additional biosignals, scaling for groups of individuals, and designing for broader applicability across various contexts, from enhancing cognitive functions and emotional comfort in elderly care, to serving as a therapeutic tool in mental health, fostering collaboration in educational and corporate settings, bridging cultural gaps, and enabling artistic collaborations.

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